HAMZA KHALID

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EDUCATION

Carnegie Mellon University (CMU), Pittsburgh, PA — Aug. 2021 – May 2023
Master of Science in Biomedical Engineering and Integrated Study in Computer Science

- **Coursework:** Principles of Software Construction: Objects, Design, and Concurrency; Data Structures; Computer Systems; Distributed Systems; Software Project Management; Computer Vision; Machine Learning; Deep Learning
- **Academic Honors:** Department Head Fellowship; Koshish Foundation Scholar.

Nadirshaw Eduljee Dinshaw (NED) University, Karachi, Pakistan — Dec. 2015 – Aug. 2019
Bachelor of Engineering in Electronic Engineering

TECHNICAL SKILLS

Programming/Web: C/C++, Python, Java, JavaScript, PHP, SQL, HTML/CSS, Django, React, Ajax, WordPress.
Frameworks/Tools/Platforms: Databricks, Docker, Apache Spark**,** Airflow, AWS, Unix/Linux, Git, REST APIs, PostgreSQL, PyTorch, Tensorflow, LangGraph, CrewAI, Pydantic AI, ML Flow.

PROFESSIONAL EXPERIENCE

Aledade, Inc. | *AI Agents, LLM, Databricks, Spark, Postgres, Software Design and Testing* — US Remote
Senior Software Engineer — June 2023 – Present

- Building and optimizing production-grade LLM applications/AI agents to automate data mapping and ingestion workflows.
- Built AI agent with vector search to fuzzy match raw data into canonical, speeding up the manual work of months into mins.
- Designed and implemented scalable raw ingestor software in Databricks with Apache Spark, streamlining data ingestion process and setting standard for future projects by integrating best practices and software engineering patterns.
- Elevated team's Databricks proficiency by leading knowledge shareout sessions, and enhancing project execution efficiency.
- Developed Databricks testing tool, optimizing testing process for development projects, and documented approach.
- Maintaining legacy system operations, ensuring team's continuous operational success.

Carnegie Mellon University | *Git, Java, JavaScript, Software Design and Testing, OOP* — Jan 2023 – May 2023
Graduate Teaching Assistant **--** Principles of Software System Construction: Objects, Design and Concurrency.

Motion2Ai, Inc. | *Python, AWS, SQL, Databases, Software Design/Testing* — Mountain View, California (Remote)
Data Scientist Intern (Software Engineering) — May 2022 – Aug. 2022

- Engineered software to visualize ~1 million miles of AI sensor data from real-world forklift operations, improving data analysis, and productivity of the data science team, resulting in a 20% increase in efficiency; deployed on AWS.
- Developed advanced data and visualization plugins using AWS S3, Athena, and PostgreSQL, incorporating scalable architecture and factory method design pattern; conducted thorough testing and quality assurance with unit tests/mocks.
- Programmed and optimized high-performance multithreaded video player using AWS Glue Crawler, boto3, and socket programming; enabled video buffering with custom cache that boosted company's video streaming speed by 400%

School of Computer Science, CMU | *Microsoft Azure, DevOps, Data Analysis* Pittsburgh, Pennsylvania
*Research Assistant*Jan. 2022 – May 2022

- Authored 3 modules covering Pandas, NumPy, and Matplotlib libraries, enabling high schoolers learn data analysis through online learning platform called Sail; collaborated with peers to troubleshoot technical issues.
- Revamped data analysis course by identifying bugs, fixing errors, and enhancing overall course quality; containerized auto-graders (Java, Python, Bash) for grading assignments using CI/CD workflows on MS Azure (docker).

Neurocomputation Lab | *Databases, Digital Ocean, Object Oriented Programming* — Karachi, Pakistan
Research Assistant (Software Engineering) — Oct. 2019 – Aug. 2021

ACADEMIC PROJECTS

Face Recognition and Verification, CMU | *Deep Learning, Computer Vision, PyTorch* — Sept 2022 – Oct 2022
• Designed and developed face classification and verification system utilizing ConvNeXt architecture, conducting research on various CNN architectures including ResNet and MobileNet, resulting in 90% accurate and efficient implementation.

MyTorch: | *Deep Learning, CNN, Numpy, Python* — Sept 2022 – Oct 2022
• Implemented minimal neural-network library using Numpy, containing classes to train feed-forward neural networks.

Data Analysis and Visualization Framework, CMU | *Java, JavaScript, GUI, REST APIs* — Nov 2021 – Dec 2021
• Led team of 2 to design Java-based framework, analyzing text with Google NLP API; created 6 data and display plugins for users to fetch, process, and visualize data; used MVC pattern for GUI (HTML/CSS/JavaScript).
• Extracted social media data in framework by Facebook, Twitter, and YouTube APIs and presented it with Chart.js.

Car's Health Monitoring and Risk | *Deep Learning, Computer Vision, PyTorch, Python* — Oct. 2019 – Aug. 2021
• Built Android app named 'OBD Charm' to fetch sensor data from cars' engine control unit for car health monitoring; used OBDII protocol for data serialization and REST APIs (PHP/SQL) hosted on Digital Ocean server for data transfer.
• Supplemented 'OBD Charm' with Google's geolocation API, Bluetooth manager, notifications service, background services,

broadcast receivers, and SQLite models with MVVM architecture; beta tested project with ~50 cars